Guanine Exchange Factors Show Existence of Transient Pockets on Rac1 Surface

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Abstract

Rac1 (Ras-related C3 botulinum toxin substrate 1) is a molecular switch that catalyses the hydrolysis of GTP to GDP. Owing to the slow nature of GTP hydrolysis and even slower dissociation of GDP, Rac1 must be assisted by Guanine Exchange Factors (GEFs) and GTPase-activating proteins (GAPs). As a result, GEFs and GAPs control the switching mechanism in Rac1. This regulatory control mechanism provides a means to modulate the Rac1 activity in several diseases in which it has been implicated. Rac1, like other small GTPases remains undruggable due to its small size, and shallow surface. However, GAPs and GEFs must binding to Rac1 to regulate its functions, and therefore, we envision presence of transient pockets on Rac2 that facilitate binding with its affectors. Hence, this work describes our attempts to locate transient binding sites on the Rac1 surface that maybe crucial for its interaction with GAPs and GEFs. We also use cosolvents as chemical probes to accelerate pocket-opening as transient pockets may require induced-fit mechanism. We found four transient sites on the RAC1 surface. Two of which form the docking site for GEFs. The third pocket is located in the region that serves as a site to interact with the DHR domain containing GEFs such as the DOCK family. And the fourth pocket is located that serves as the interaction site for Engulfment

And Cell Motility 1 (ELMO1) protein that assists DOCK family of GEFs to perform their function. In the absence of GEFs, the pocket volumes are too small to be druggable, indicating their transient nature. Using the pocket volume fluctuations, we evaluate their relevance for targeting with small molecules. Following co-solvent based molecular dynamics, the principal component analysis suggests that most of the pocket opening and closing is dictated by the switch regions movements.

Introduction

Small GTPases are a ubiquitous class of proteins that play pivotal roles in various cellular processes. Their functions include signal transduction, vesicle trafficking, cytoskeletal dynamics, and cell division. Ras GTPases regulate cell proliferation, differentiation, and survival. Rho GTPases govern cytoskeletal dynamics and cell migration. These proteins possess conserved domains across several families; made up of six β-sheets surrounded by 5 α-helices. All domains are conserved such that they carry a guanine nucleoside binding site to recognise the guanine base and phosphate groups and the magnesium ion.[1,2] The switch regions regulate the traffic to the guanine nucleoside binding site. Their activity is regulated through a molecular switch mechanism by guanine nucleotide exchange factors (GEFs), GTPase-activating proteins (GAPs), and guanine nucleotide dissociation inhibitors (GDIs).[2] The mechanism cycles between OFF (GDP-bound inactive) and ON (GTP-bound active) states. This switching mechanism controls various processes like cell growth and differentiation to vesicular and nuclear transport.[2,3] It is also well established that GTPases alone cannot hydrolyse GTP effectively. Therefore, GEFs and GAPs are known to assist in the catalytic function, and thus the switching mechanism. They control the dynamics of the switch regions that facilitate either the hydrolysis of GTP or the dissociation of GDP. GDP:Mg^{2+} dissociates from GTPases very slowly, and therefore, GEFs assist in exchanging GDP for GTP. One of the accepted mechanisms of GEFs in the dissociation of GDP is by electrostatic repulsion of the magnesium ion that forms a strong interaction with GTPases and phosphate groups of GDP. The loss of a magnesium ion weakens the affinity of GDP, which eventually dissociates. Alternatively, some GEFs can rearrange the switch I or II conformations to reduce GDP's affinity, thus facilitating its dissociation. However, GEFs can also combine the two mechanisms to enable GDP dissociation.[4,5] Furthermore, GAPs stabilise the transition state for GTP:Mg^{2+}:GTPases, and facilitate the hydrolysis of the γ-phosphate group from GTP. Taken together, the switching mechanism can be seen as GEFs assist in the GTPase activation, while GAPs assist in GTP hydrolysis. Following GTP hydrolysis, the inactive GTPases prevent further signal transduction.[6] In some classes of GTPases, the active and inactive states are conjugated with GTPases being shuttled between the cell membrane and cytosol. Such GTPases undergo post-translational modifications to carry long-chain fatty acids at their C-terminal to assist bilayer insertion. GDIs[7,8] sequester the lipid component to prevent membrane insertion of GTPases in the cytosol. This coordinated action of GTPases, GEFs, GAPs, and GDIs is essential for various cellular processes. Dysregulation of these components contributes to pathologies like cancer, neurodegeneration, and inflammatory disorders. For instance, mutations in Ras and its regulators are implicated in oncogenesis, highlighting the therapeutic potential of targeting GTPase signalling pathways.[2] Such mutations (Q61L, Y72C, oncogenic P29S and Q61R, and negative T17N) are known to disrupt the equilibrium between the active and inactive states of GTPases.[9] It has been observed that mutations either enhance the affinity of GEFs or GAPs that trap the small GTPases either in an activated state or an inactivated state. Small molecule inhibitors targeting GTPase activity or disruptors of GEF/GAP interactions[10] are under development for cancer therapy. However, there are several challenges in targeting GTPases and/or GTPases-GAP/GEF complexes. Firstly, targeting the guanine binding site is challenging owing to the large intracellular concentration of GTP and high affinity.[11] In such cases, finding alternate sites can avoid such competition with endogenous substrates. GTPase hyper-activation or inactivation can be corrected by disrupting the protein-protein interaction (PPI) between GTPases and GEFs/GAPs by targeting the regulatory protein docking sites.

Rac1 plays a crucial role in regulating cellular processes owing to its switching mechanism that recruits several proteins upstream and downstream. PREX1, TIAM1, VAV1 and DOCK2 are Rac1-specific GEFs that regulate many cellular processes and, therefore, are implicated in many Rac-dependent pathologies such as asthma and cancer. Structurally, PREX1, TIAM1 and VAV1 are DH/PH domain-containing proteins, while DOCK2 contains a DHR domain. Irrespective of their structural diversity, GEFs interact with the switch I and/or II region residues. Since the nucleotide-binding site is well conserved among the GTPase family, this poses a significant challenge to selectively target Rac1 by modulating the nucleotide binding site. Furthermore, the molar concentration with picomolar binding affinity of GTP make it more difficult for small molecules to competitively inhibit Rac1. The binding of GEFs with Rac1 shows the possibility of alternative binding sites that transiently open to allow this protein-protein intraction. Therefore, the switching mechanism offers a conformational landscape that must be studied in depth to locate other binding sites on the Rac1 surface. Therefore, we embarked to understand the complete dynamical landscape of Rac1 protein bound to either nucleotides (GDP or GTP) or GEFs (PREX1, TIAM1, VAV1 or DOCK2). In this study, we determined the evolution of Rac1 binding pockets in its nucleotide-bound conformation and bound with its partners (holo). Following molecular dynamics simulations, we identified and measured the pocket volume changes for the transient pockets. We also monitored the effect of binding of various GEFs and nucleotide on the changes in the pocket volume. Lastly, using pearson's correlation coefficient, we evaluated the possibility for correlation between changes in all four transient pockets among each other.

Methods

Preparation of systems for molecular dynamics simulation

The coordinates for all complexes were retrieved from the Protein Data Bank (PDB); the PDB entries used in this study are listed in **Table 1**. The X-ray structures were imported and processed in the UCSF Chimera v1.17.[12] Any missing loops were built using Modeller[13] with UCSF Chimera v1.17 as the GUI.[14,15]

Table 1: Rac1 conformations and binding partners used in this study

PDB id	Rac1 Modulator/Ligand	Comments
1HE1[16]	GDP:Mg^{2+}	Inactive Rac1
3TH5[17]	GTP:Mg^{2+}	Active Rac1
4YON[18]	PREX1 (aa. 34-366)	DBL family member
1FOE[19]	TIAM1 (aa. 1033-1406, mouse)	DBL family member
2VRW[20]	VAV1 (aa. 170-575, mouse)	DBL family member
2YIN[21]	DOCK2 (aa. 1192-1622)	DOCK family member

Generating AMBER-compatible parameters

All systems were prepared for MD simulations with the Leap module in AMBERtools23. Protein atoms were atom-typed according to the AMBER14 (ff14SB) force field.[22] Amber-compatible parameters for all nucleotides were obtained from the Manchester AMBER database. For the VAV1 zinc finger region, the parameters of the Zinc atom, coordinating cysteines and histidine residues, the force field parameters were obtained from the ZAFF[23] force field library.

Molecular dynamics simulations

All complexes were solvated with TIP3P[24] waters enclosed in a box with dimensions extending 10Å beyond any protein atom, and an appropriate number of neutralising ions were added to maintain the system's neutrality. The complexes were relaxed using three cycles of minimisation of 50000 steps each. The first 5000 steps of minimisation adopted the steepest descent algorithm, and the remaining 45000 steps used the conjugate gradient algorithm. The cut-off for evaluating non-bonded interactions was placed at 8.0Å for the minimisation phase. The first cycle of minimization was performed with 25 kcal/(molÅ^2) restraining force on the solute atoms, this force constant was reduced to 5 kcal/(molÅ^2) in the next cycle, and the final cycle was performed without any restraining forces on either the solute or the solvent atoms. The system was heated in 4 stages (see **Table 2**) from 0 K to 300 K for 10 ns with 50 kcal/(molÅ^2) restraining force on the solute atoms using the Langevin dynamics, with a collisional frequency of 2 ps^{-1}. The next step employed the NPT ensemble for density equilibration for another 2 cycles of 10 ns each, with the restraining force on the solute atoms gradually decreasing from 5 to 0 kcal/(molÅ^2). In the equilibration phase, the cut-off for evaluating non-bonded interactions was placed at 8 Å. In the production phase of the MD simulations, all systems were simulated for 100 ns in triplicate, amounting to a 300 ns trajectory for each system. The long-range electrostatic interactions were calculated using the Particle Mesh Ewald (PME) summation method.[25] The cut-off for evaluating non-bonded interactions was placed at 9.0 Å for the production phase. All the computations were done using the CUDA-enabled PMEMD code[26–28] in AMBER22.[29,30] All simulations were performed with the SHAKE algorithm[31] to constrain bonds to hydrogen atoms, with the tolerance set to 0.00001 Å. A 2 fs integration time step was used to solve Newton's equations of motion. For analysis, the conformations were saved every 100 ps, amounting to 1000 snapshots per trajectory (3000 per system).

Table 2: Details of stages of Heating using NVT Ensemble

Stage	Temperature range (K)	Time (ns)
Stage 1	0 to 100	0 to 2.5
Stage 2	100 to 200	2.5 to 5
Stage 3	200 to 300	5 to 7.5
Stage 4	300 to 300	7.5 to 10

Measuring the structural dynamics of Rac1

Analysis of the MD trajectory was performed with the cpptraj[32] module in AMBER-Tools23. Before the analysis, all solvent atoms and neutralising ions were stripped off, and the protein conformations were aligned to their respective X-ray conformation. The mass-weighted Cα root-mean-square deviation (RMSD) for the protein was computed using the X-ray conformation as the reference. The root-mean-square fluctuations (RMSF) per residue for the backbone and side-chain atoms were computed to understand the domain-specific fluctuations.

Identification and measurement of alternate pockets on Rac1

POVME2[33,34] was employed for identifying various pockets on the Rac1. To be able to incorporate various experimental conformations of Rac1, we employed Rac1 with different binding partners (see **Table 1**). To identify the pockets, the pocketID module on POVME2 was used on the aforementioned Rac1 structures. The parameters used for pocket identification are shown in **Table 3**. Following the pocket identification, we monitored the pocket volume changes of these pockets throughout MD trajectories. It has been observed that the pocket might be a good candidate for druggability if its volume range was found to lie between 610Å^3 to 930Å^3.[35–37]

Table 3: Parameters used for pocket identification

Parameter	Value
pocket detection resolution	4.0 Å
pocket measuring resolution	1.0 Å
clashing cutoff	3.5 Å
number of neighbors	4
number of spheres	5
sphere padding	5.0 Å

MD simulations with Cosolvent probes

MD simulations with cosolvent probes are very useful to accelerate pocket opening or to keep the transient pockets open for a longer timescale in MD simulations to better understand pocket characteristics. We identified transient pockets in response to GEF binding, which were otherwise not observed or very hard to locate. To ensure that these transient pockets can remain open sufficiently long to be druggable, we used cosolvent MD simulations. For this purpose, we used 1M concentration of isopropanol, phenol and toluene. The choice of these solvents mimics drug-like fragments with decreasing water solubility. In this case isopropanol (logP = 0.05) > phenol (logP = 1.39) > toluene (logP = 2.56). The simulation box with Rac1 protein (PDB: 3TH5, GTP and Magnesium ions were deleted) and 0.15M of NaCl was prepared using packmol.[38] The protocol for MD simulations is the same as mentioned above. GAFF2 parameters were assigned to the organic solvents. For cosolvent MD simulations, we performed triplicate MD simulations for 1 μs, amounting to a total sampling time of 3 μs per solvent system. For the baseline calculations, we also prepared two systems without adding cosolvents. One is the RAC1 complexed with GTP and Magnesium ions, which mimics an active-like conformation of

RAC1. The apo system was created by deleting GTP and Magnesium ions from the active site system described previously. These systems would enable us to understand if any of the transient pockets can spontaneously open without the influence of the cosolvents and any ligand in the nucleotide pocket.

Principal component analysis of Rac1 protein following cosolvent MD

To understand the effect of cosolvent probes on the Rac1, and thus opening of the transient pockets, we extracted the essential dynamics using Principal Component Analysis (PCA). The Variance-Covariance matrix for Cα atoms was extracted in the bio3D package[39] module in R Studio. We extracted and analysed four principal components; however, the first principal components with the largest eigenvalues was used to understand the essential dynamics in the Rac1 protein. The scree plot for each system is given the supplementary figure S3

Quantifying the GEF binding hotspots on Rac1 surface

The free energy of binding was computed using the equation (Eq-1) on the conformations saved from the MD simulations using the MMPBSA module[40] available in AMBER-Tools23;

$$\Delta G_{\text{binding}} = \Delta E_{\text{MM}} + \Delta H_{\text{GBSA}} - T\Delta S \tag{1}$$

Where ΔE_{MM} is the interaction energy between the binding partners computed in the gas phase using a molecular mechanics method. The solvation-free energy or the enthalpy of hydration ΔH_{GBSA} was computed using the Generalised-Born Surface Area (GBSA) method, and the entropic component of binding $T\Delta S$ was not computed due to the huge computation cost. The internal dielectric constant for GBSA calculations was set to 1, and the solvent dielectric constant was fixed at 80. For computing the free energy of binding using the MM-GBSA method, the OBC's GB model[41] ($igb = 5$)

was used. The mbondi2 radii set was employed to define the PB radii for all solute atoms during the preparation of the topology and parameter files in the leap module in AMBERTools23. The ionic strength for GBSA calculations was maintained at 150 mM. All energy components, in an implicit solvent, were calculated using a large cut-off of 999 Å. The surface area for the non-polar solvation term was computed using the linear combination of pairwise overlaps (LCPO) method.[42] To identify the critical amino acid residues that are critical for the protein-protein interactions between Rac1 and GEFs, we computed the per-residue decomposition of the binding affinity value. This will also enable us to estimate the contribution from the residues lining the newly identified transient pocket and their energetic contribution to GEF binding.

Results and discussion

Protein-protein interactions are central to biology. By coordinating an intricate network of proteins that are recruited and activated in a pathway to pass the signal to the nucleus to control the rate and amount of protein expression. Such networks are tightly regulated by molecular switches, majorly by small GTPases, which are activated in response to GTP binding and turned off upon GTP hydrolysis to GDP. For the next switch on step, GDP must dissociate, which allows for GTP binding. However, the binding affinity of GDP and GTP are very high and comparable, therefore, GDP dissociation is not spontaneous. Thus the guanine exchange step must be assisted by GEFs that are recruited depending on the pathway. The exchange factors serve a dual role of nucleotide exchange, and recruiting proteins downstream in the signal pathway by activating a specific GEF-deppendent network of proteins. Therefore, we attempt to understand the Rac1 dynamics bound to different GEFs, nucleotides, and *apo* states. Further, the Rac1-GEF binding paves way to unlock different binding sites on Rac1 surface that are not normally open in the *apo* state. We further investigate whether such sites are amenable to small molecule targeting by measuring the pocket volume changes over the course of MD trajectories. We also try to explain the structural changes that Rac1 undergoes during the opening of these alternate sites, and if there is any noticeable change in the nucleotide binding site that may effect the GTP association or GDP dissociation.

GEF docking sites open transient pockets on the Rac1 surface

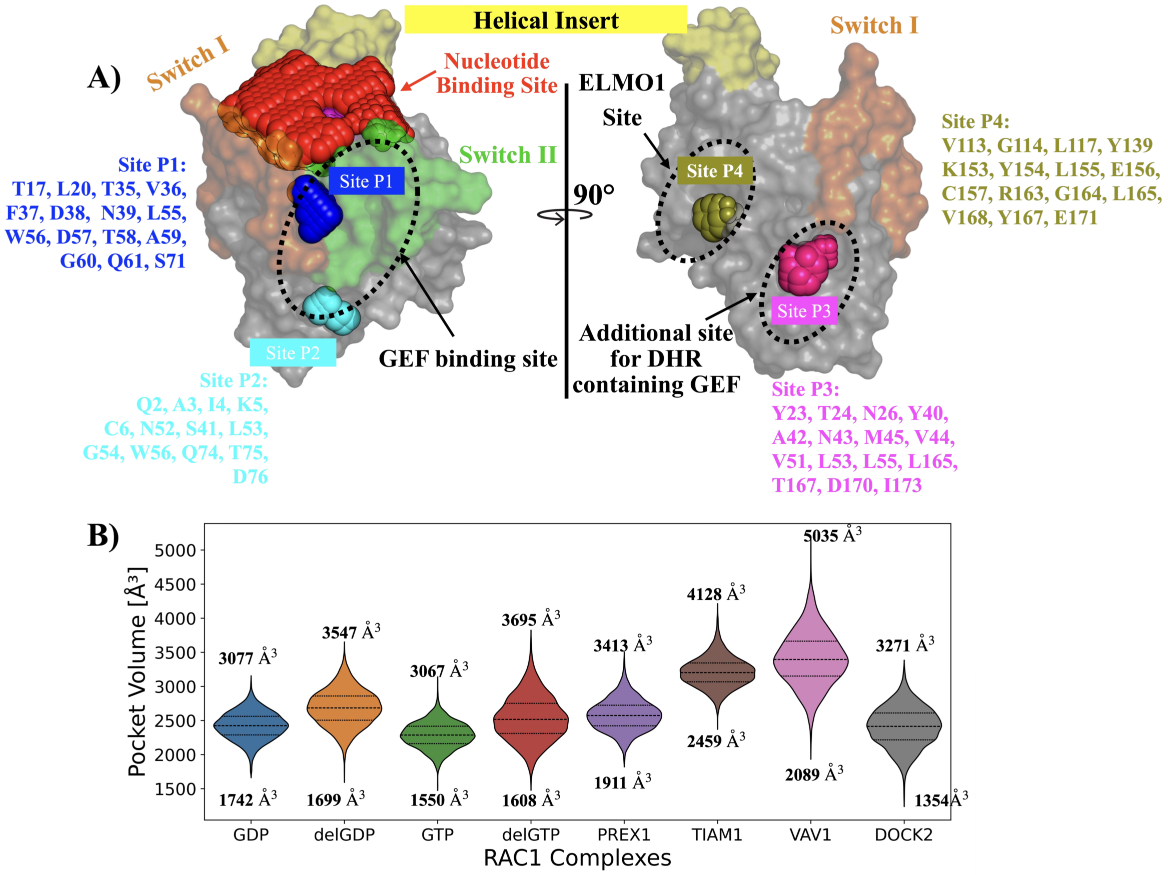

Figure 1: **A**) Rac1 surface with all possible binding pockets. The yellow-coloured region depicts the helical insert, which is peculiar to the Rho GTPase family. The orange-colored region depicts switch I, and the green colored region is switch II. Red spheres occupy the nucleotide-binding site. The pink-colored region is the P-loop. Blue spheres depict Site P1, cyan spheres show Site P2. Magenta spheres occupy the Site P3, and asparagus spheres show the Site P4. **B**) The violin plot illustrates the distribution of the pocket volumes for each of the nucleotide-binding sites. The changes in the pocket volumes are monitored when Rac1 is complexed with either GEFs, GDP, GTP or in the *apo*-states named as delGDP (GDP deleted from GDP-bound system), and delGTP (GDP deleted from GTP-bound system). The amino acid numbering for Rac1 follows the uniprot id: P63000. The numerical values on the violin plots indicate the minimum and maximum pockets volumes measured in the MD trajectories.

We employed the POVME2[34] tool to identify and measure the pocket volumes of new transient sites. By definintion transient or cryptic pockets are observed only in holo proteins, while being undetectable in apo proteins. Therefore, we used various Rac1 structures bound to GEFs and nucleotide (**Table 1**). To measure the pocket volume changes of these pockets, we simulated all Rac1 complexes(**Table 1**). Before measuring the pockets volumes, all binding partners, including nucleotides, were deleted from the trajectory. In other words, the Rac1 comformation was allowed evolve over time in presence of the binding partners, which allows opening of the transient sites. Firstly, we observed the change in the nucleotide pocket volume influenced by GEFs, and the nucleotides. We observed that the pocket volume fluctuations are remained under 3000Å^3 in the presence of GTP, and GDP (**Figure 1B**). This indicates the closing of the nucleotide pocket in preparation for GTP hydrolysis (GTP-bound). For GDP-bound structure, this would indicate a state after the GTP hydrolysis. The maximum and minimum pockets volumes for nucleotide site are similar for both GTP- and GDP-bound Rac1. Next, we monitored the changes in the pocket volumes by deleting the GTP and GDP to create *apo*-like states called "delGTP" and "delGDP" respectively, to understand the opening of the nucleotide pockets without the influence of GEFs. As expected, we noted a significant increase in the maximum pocket volumes over 3000Å^3 for both the *apo*-like states, while the minimum pocket volumes stayed close to those observed nucleotide-bound states This marked increase in the pocket volumes in the nucleotide-free state can be attributed to an increased flexibility in the switch regions and the P-loop (**Figure 2B** and **supplementary figure S6**), which are directly interact with the nucleotides. A more noticeable difference in flexibility(**Figure 2B**) can be seen in the P-loop and the switch I loop (**supplementary figure S6** and **S7**).

The switch region conformation is also found to be reduced by GEF binding (**Figure 2C**), suggesting their role in controlling the nucleotide traffic by modulating the switching regions. We tested this hypothesis by simulating a nucleotide-free and GEF-free form of Rac1 (delGDP and delGTP in **Figure 1B**) and found that the mean pocket volume and fluctuations showed a greater tendency to increase, however, only marginally, as maximum volume was still under 4000Å^3. Next, we measured the pocket volume and the atomic fluctuations in the presence of GEFs in the nucleotide-free state. The volume fluctuations are higher than in the nucleotide-free state without GEFs. However, the volume changes showed significant changes among different GEFs. For example, PREX1 and

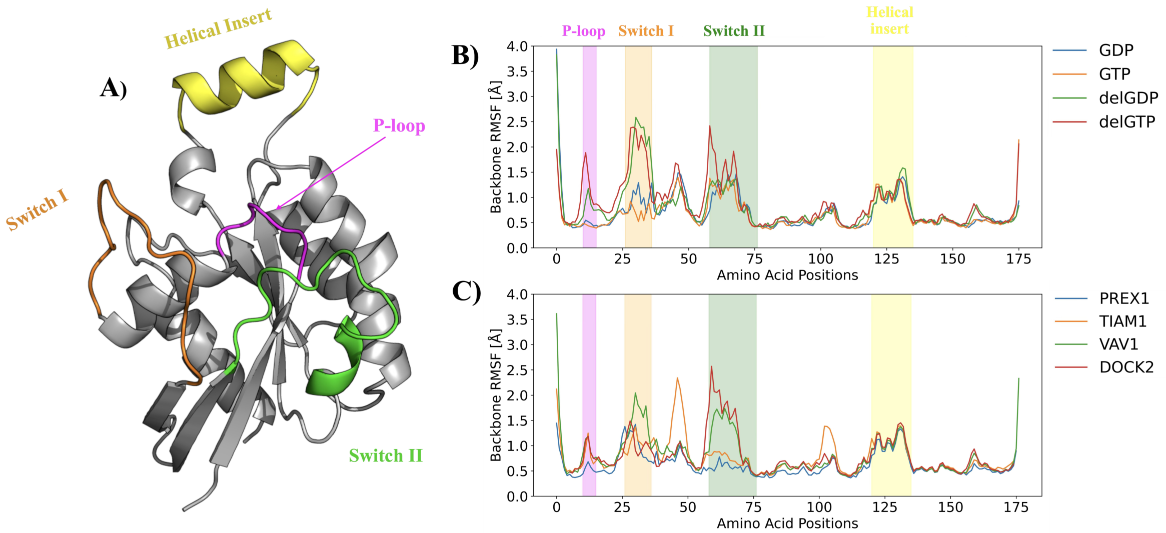

Figure 2: Rac1 domains and dynamics of each region. **A)** A Cartoon representation of human Rac1 (**PDB id:1HE1**). The P-loop region is depicted in the magenta colour. The Switch I region is depicted in Orange Colour, and the Green coloured region is the Switch II region. The helical region is shown in yellow colour. **B)** Atomic fluctuations of the backbone atoms are measured as root-mean-squared fluctuations (RMSF) for the human Rac1 domain to compare the effect of nucleotide binding on the dynamics of various domains. **delGDP** and **delGTP** are systems created by deleting the nucleotides from the Rac1 binding pocket to create corresponding *apo* system. **C)** Atomic fluctuations of the backbone atoms are measured as root-mean-squared fluctuations (RMSF) for the human Rac1 domain to compare the effect of GEF binding on the dynamics of various domains.

DOCK2 showed volume changes that are similar in magnitude to *apo*-like states (delGDP and delGTP in **Figure 1B**). TIAM1 and VAV1 showed significantly enlarged nucleotide pockets with maximum volume above 4000Å^3. We hypothesise that such a difference in pocket volumes by different GEFs suggests that the X-ray structure has captured the Rac1-GEF complexes either just after GDP dissociation (PREX1 and DOCK2) or in a state that was pre-organised for GTP:Mg^{2+} association (TIAM1 and VAV1). As we have already established that pocket volume changes in the nucloetide can be attribute the P-loop and switch region dynamics, it becomes apparent that GEFs interacting with primarily the switch region can control the nucleotide exchange (**figure 2C**). The nucleotide is seen to enlarges with all GEFs that is pre-requisite for eject GDP, and then perhaps enlarge even more to allow association of GTP and Mg^{2+}.

We identified four transient pockets on the Rac1 surface that cannot be located with Rac1 structure alone. The sites P1 and P2 are located in the region lining the switch regions I and II. These sites form the binding site for the GEFs that perturbs the switch

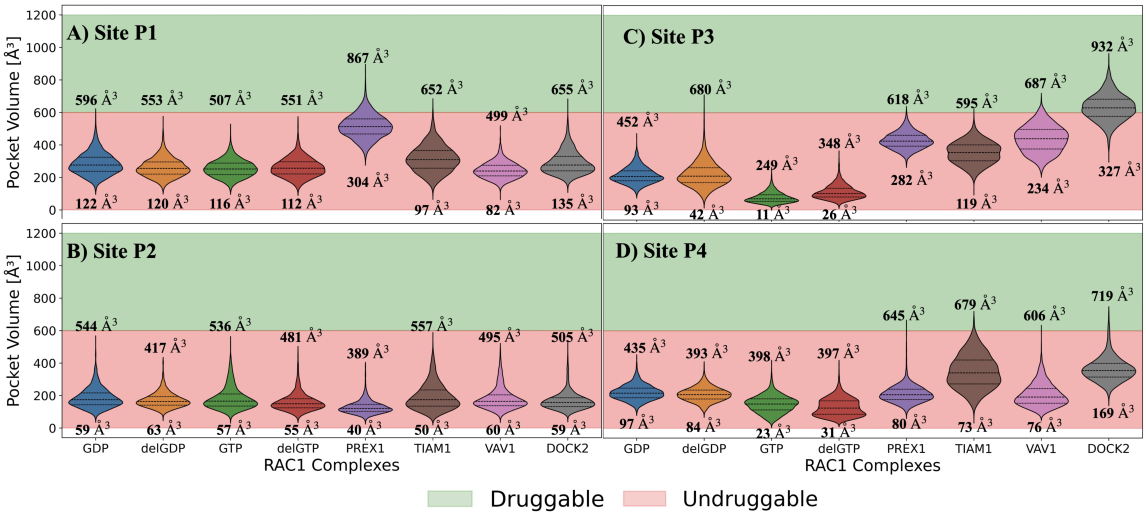

Figure 3: Four binding sites identified using POVME2. **A)** Site P1 is located between the switch I and II. It appears to be the site similar to KRAS-G12C covalent inhibitors. **B)** Site P2 is located at the bottom of the switch I and II loops at the N-terminal region. **C)** Site P3 is located at the site from where the switch I region. **D)** Site P4 is the only pocket that is not located close to switch I or II regions, however, this site appears to accomodate ELMO1 protein that assist DOCK family of GEFs in their function. The threshold for druggability of a pocket was set according to the following references.[35–37] The numerical values on the violin plots indicate the minimum and maximum pockets volumes measured in the MD trajectories.

regions leading to opening of these sites. Our pocket volume measurments indicate that the volumes are large enough to accommodate small molecule binders. In the absence of the GEFs, these sites are too small, and hence, not detected in GEF-free Rac1(**Figure 1A**). Site P1 in Rac1 is similar to the one in K-RAS and H-RAS[43,44] that is occupied by the covalent inhibitors in G12C Ras mutant. Sites P1 and P2 are present at a crucial location that could modulate the dynamics of the switch regions. Thus, we hypothesise that one of the consequences of targeting these sites is that it could compete with GEF binding. Alternatively, the molecules occupying these sites could act as molecular glues that stabilise the Rac1-GEF complex slowing the switching mechanism, and thus, the nucleotide exchange. Both the scenarios suggest that the modulation perturbs the GEF-mediated downstream signalling (**Figure 3B**). Due to the differences in the binding volumes caused by different GEFs, we anticipate that different Rac1-GEF complex could be selectively targetted by small molecules of different binding complementarity. We also acknowlegde that one of the secondary effects of targeting site P1 and P2 opens up a possibility to occlude GTP binding, albeit non-competitively. According to one of the accepted mechanisms of nucleotide exchange,[2] GTP binding weakens the GEF binding, which eventually dissociates. By preventing or decreasing the rate of GEF dissociation, GTP binding could be prevented or slowed down. Thereby prolonging the nucleotide-free state and preventing the downstream signalling. Site P2 is located below Site P1, and also found to marginal increase in the pocket volume in the presence of GEFs (**Figure 3C**). We propose that, owing to the proximity, this could serve as an extension for molecules binding to Site P1 (**Figure1A** and **supplementary figures S4A and S4B**).

Away from the GEF binding site are located sites P3 and P4 (**Figure 1A**). Site P3 is located towards the N-terminal side of the switch I region. From the X-ray crystallographic informations, Residues lining the site P3 are known to interact more specifically with DHR-containing GEFs belonging to the DOCK family.[21,45]Targeting this site could modulate the switch I region dynamics, which in turn could have an influence on the traffic to the nucleotide-binding site as well as GEF binding. The ELMO1 protein that facilitates the binding of the DOCK family of GEFs is known to interact with residues that form the site P4.[45] Taken together, we hypothesise that modulation of site P3 and P4 could specifically target and modulate the downstream signalling involving DOCK family of GEFs.

Accelerating the pocket opening using cosolvents

Transient pockets open in response to ligand binding, suggesting an induced fit mechanism. However, in the absence of known binders, cosolvents that structurally resemble drug-like fragments are used as chemical probes to accelerate the opening of binding sites that require ligand binding. In this work, we used three commonly used cosolvents with different hydrophobicities to assess that ability of these transient sites to open up in the absence of GEFs suggesting their amenbility to be targeted by small molecules. The PBD entry with accession number 3TH5 is the only X-ray crystal structure that is crystallised in the active-like conformation bound to GTP analogue, Guanylyl imidodiphosphate (GNP), without any regulatory protein. Rac1 conformation in this case shows that all GEF induced pockets are in the undruggable state evident by thier small pocket vol-

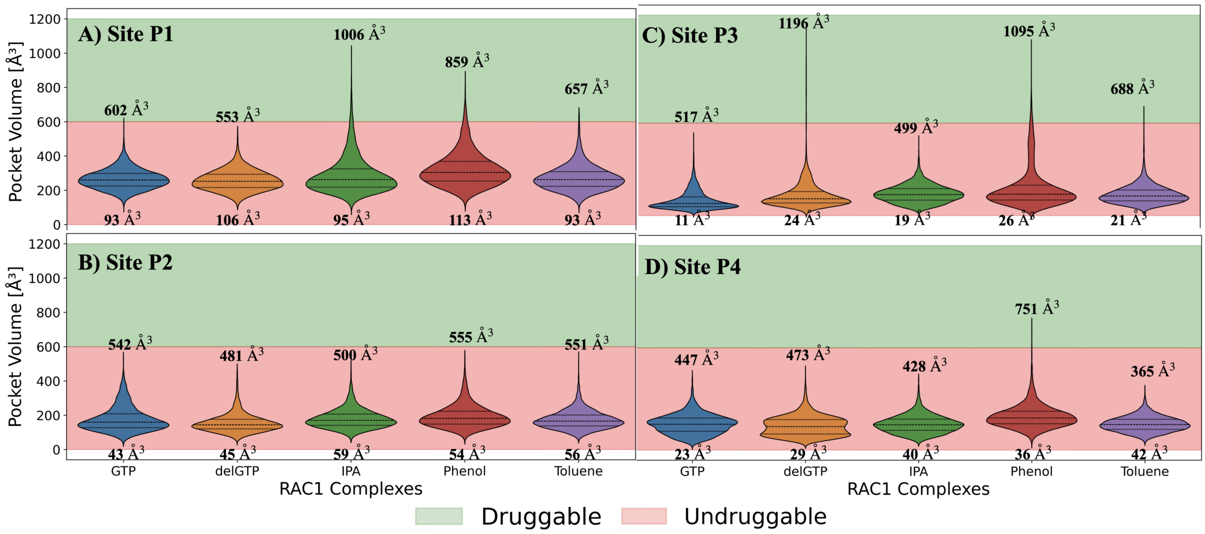

Figure 4: Pocket volume measurements following cosolvent MD simulations for **A)** Site P1, **A)** Site P2, **A)** Site P3, and **A)** Site P4. The numerical values on the violin plots indicate the minimum and maximum pockets volumes measured in the MD trajectories.

umes. We performed microsecond timescale MD simulations to check the effectiveness of the chemical probes to open the transient pockets through induced-fit mechanism. We observe that three sites, P1, P3 and P4, the pocket volumes cross the threshold value. Thesefore, suggesting that these pockets can open large enough to be targeted by small molecules. However, systems (GTP and delGTP) without cosolvent probes remain below the threshold (**Figure 4**) value confirming their transient nature. Thus, warrants an induced-fit mechanism that is involved in open of these pockets. Physiologically this induced-fit mechanism is observed in response to receptor pre-organisation step for GEF binding. In case of site P1, the pocket volume for the reference systems (GTP and delGTP) largely remains below the threshold for druggability. The chemical probes, isopropanol and phenol, are able to open this site significantly larger than the reference system attaining a maximum volume of 1006Å^3 and 859Å^3 respectively(**Figure 4A**). While using a completely hydophobic probe, toluene, the pocket volume increased only marginally (maximum volume measured is 657Å^3). This can be attributed to the phase separation observed for toluene molecules due to its high hybrophobicity which leads to practically insoluble in water. To prevent such phase separation, it is recommended to use intermolecular repulsive potential applied to probe molecules.[46] For site P2, none of the chemical probes were able to increase the pocket volume above the threshold value (**Figure 4B**). Upon careful observation, we note that the site P2 is lined by β-sheets which forms majority of the site. This leads to a rather flat pocket lined by rigid secondary structural elements living little scope for movements that could increase the pocket volume. However, we expect this site P2 to serve as subpocket for site P1 site binders that can be expanded to site P2 to increase the binding affinity.

Site P3 appears to be open spontaneously in *apo*-like (delGTP) with a maximum volume measured is 1196Å^3 (**Figure 4C**). While the pocket volume remained beyonf the threshold for GTP-bound Rac1, and with IPA cosolvent. However, phenol and toluene could push the maximum volume beyond the threshold, wherein with phenol the maximum volume measured was 1095Å^3 which is similar magnitude to delGTP. The residues that make up site P3 appear to be majorly aromatic and aliphatic that makes this pocket a hydrophobic patch (**Figure 1A**). Therefore, the hydrophilic cosolvent, IPA, was unable to open the pocket volume, whereas the hydrophobic probes like phenol and toluene did open the pocket way beyond the druggability threshold. Site P4 appears to be smallest among the four transient pockets identified in this work. We observed that only in phenol cosolvent system the pocket volume crossed the druggability threshold with maximum volume of 751Å^3(**Figure 4D**). Upon closer inspection of the binding site residues, we observe a good balance of hydrophobic and charged residues owing to which phenol, which comprises of a hydrophobic ring with hydrophilic hydroxylic group, was able to open to the pocket beyond the threshold value. In summary, we note that three out of four pockets, are indeed transient in nature, which can be open in an induced-fit manner. This is exemplified by the GEF-binding, and the cosolvent probe molecules. We did choose GTP-bound conformation of the Rac1 protein in which all the four pockets reported the pocket volumes below the threshold value(**Figure 3**). This way we could assure that in the starting conformation all pockets could be labelled as practically undruggable, giving an unabiased starting to test the ability of the cosolvent probes to open the pockets.

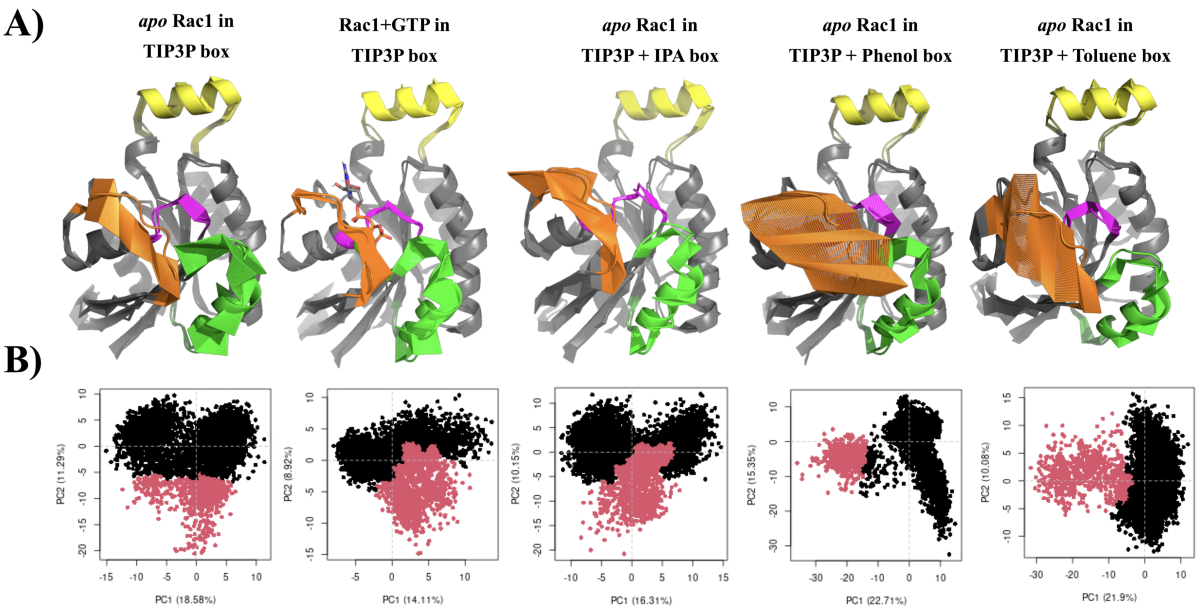

Figure 5: A) Essential Dynamics of Rac1 protein to explain the opening and closing of the transient pockets induced by the cosolvent probes. Magenta colour indicates the P-loop region. Orange colour indicates switch I region. Green colour indicates switch II region. Yellow colour indicates helical insert. IPA is isopropanol. **B**) Identification of Rac1 conformationaal clusters in various water and cosolvent systems based on essential dynamics using first two PCs..

Switch regions contribute for the maximum flexibility in the Rac1 dynamics

To understand the protein dynamics that are involved in opening of these transient pockets, we extracted the essential dynamics using the principal component analysis of Rac1 following microsecond scale cosolvent MD simulations. The purpose of such long timescale simulations with and without cosolvents is to evluate the possiblity of spontaneous opening of the pockets, and to confirm the induced-fit mechanism needed for opening the pockets. The systems that were simulated without cosolvents are treated as the reference systems for comparing the cosolvent induced pocket opening. In RAC1, like all other small GTPases, the switch regions shows higher conformational flexibility. This flexibility is central to the switching mechanism for small GTPases that allows GEF, and GAP binding, which is pre-requisite for nucleotide exchange and GTP hydrolysis respectively. The essential dynamics extracted from the dominant modes of motions by employing principal component analysis also confirms that large dynamical flexibility of the switch regions. Therefore, we embarked on understanding how the switch region dynamics could play a role on the opening and closing of these transient pockets. GTP and delGTP systems that were simulated without cosolvent show that switch regions show maximum movements in the Rac1 protein (**Figure 5A**). A significant ordering of switch I, and the P-loop is observed in presence of GTP and Mg^{2+} ion. This is attributed to the strong electrostatic interactions shown by the phosphate group through the Mg^{2+} ion which interacts with the switch I residues. The Rac1 system simulated on the cosolvent box with nucleotides shows the largest dynamics in the switch I region (**Figure 5A**), while the maximum pocket volumes are also observed in these conditions(**Figure 4**). We do not observe any other structural domain in Rac1 with dynamical flexibility as large as the switch I region. As we compare the effectiveness of the cosolvent probes in opening the transient pockets, phenol and toluene show maximal efficiency in which the PCA-based clustering (**Figure 5B**) using the first two principal components shows distinguishable conformations between open and closed states. Since phenol and toluene are fragement that frequently appear in drug-like molecules, they are more efficient in exhibiting induced-fit opening of the pockets, a behaviour that can be expected from drug-like molecules.

Per residue decomposition of analysis Rac1 residues lining the four transient pockets

So far we have observed that most of the transient pockets in Rac1 open to accommodate the GEFs required for the nucleotide exchange. Hence, we found it prudent to analyse the energetic contribution of the Rac1 residues that are involved in GEF binding. Furthermore, we attempt to locate how many amino acid residues important for GEF binding also make up the four transient sites, and thus, the per residue contribution can be indicate the binding hot spots for small molecules. As described earlier, there is a dominant role of the switch regions in GEF binding that leads to dissociation of GDP and association of GTP. This nucleotide exchange, catalysed by the GEF, necessitates their interaction with the switch regions, thus modulating their conformation such that it facilitates this exchange. The sites P1, P2 and P3 are located at the site where GEFs bind to Rac1. A thorough analysis for the X-ray crystallographic structures suggests that important amino acid residues in Rac1 involved in GEF binding and activation are located in the switch I and II regions.[18–21]

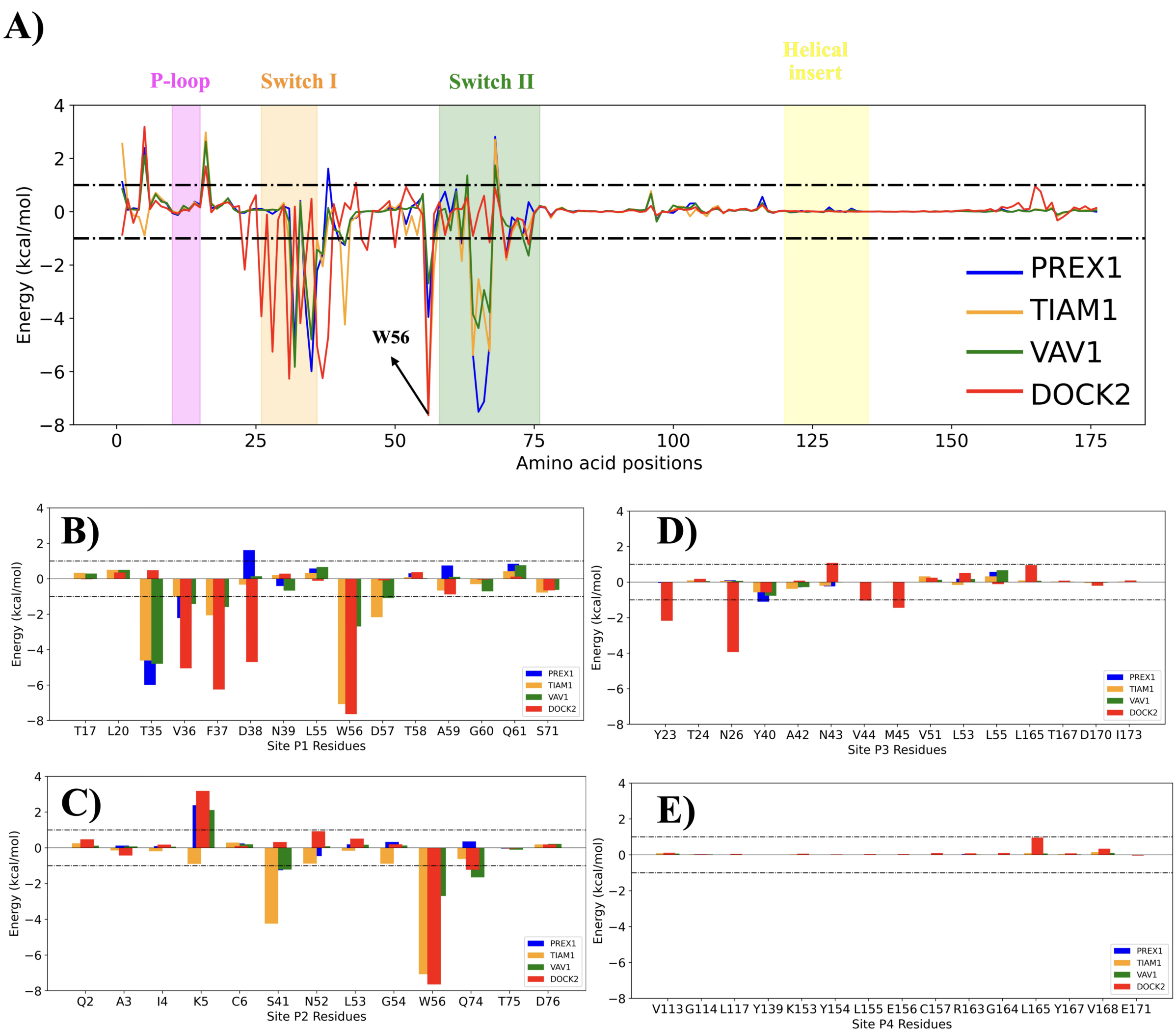

Figure 6: **A)** Per residue contribution to the effective binding affinity of the Rac1 residues to the GEFs as computed by MMGBSA. The P-loop region is depicted in the magenta colour. The Switch I region is depicted in Orange Colour, and the Green coloured region is the Switch II region. The helical region is shown in yellow colour. The residue W56 is marked due to its biological importance in the GEF binding **B)** Per residue contribution from Site P1 residues. **C)** Per residue contribution from Site P2 residues. **D)** Per residue contribution from Site P3 residues. **E)** Per residue contribution from Site P4 residues. The "**dot-dash**" line indicates an arbitrary cutoff placed at **±1 kcal/mol**, which suggests that energy values outside this region contribute significantly to the binding of GEFs.

Switch I region

Ala-27: This residue is a key determinant of specificity for DOCK GEFs. The conformation of switch 1 of Rac1 when bound to DOCK2DHR2 is compatible with Alanine (Ala) at position 27; larger side chains at this position, such as Lys-27 (found in Cdc42), would cause intramolecular steric clashes, preventing DOCK2 from activating Cdc42. Phe-28: Movement of this residue promotes nucleotide release by removing it from interactions with the guanine moiety of the nucleotide. Tyr-32 The hydroxyl group of Tyr-32Rac1 forms hydrogen bonds with conserved GEF residues, such as Glu-56P-Rex1 and Glu 201Vav1. In the DOCK2-Rac1 complex, Tyr-32R maintains its hydrogen bond to Cys-18R, unlike Cdc42 in complex with DOCK9. Thr-35 and Val-36: The backbone amides of these residues form hydrogen bonds with conserved GEF residues, such as Glu-56P-Rex1 and Glu 201Vav1. The conformational change induced by P-Rex1 binding displaces Thr-35 approximately 6 Å toward P-Rex1, thereby disrupting the ability of Rac1 to coordinate Mg^{2+}. Thr-35 and Val-36 are also important contact residues in the Tiam1 interface. Ile-33 This is one of the six divergent residues that determines specificity between Rac1 and Cdc42 for DOCK GEFs. The bulkier Ile-33R in Rac1, when compared to Val-33C in Cdc42, contributes to unfavorable binding/steric clashes with Cdc42-specific GEFs like DOCK9DHR2. Asn-39: This residue forms hydrogen bonds with GEF residues in the CR3 region (e.g., Gln-197P-Rex1 in P-Rex1). It is also one of the six divergent residues determining DOCK specificity.

Switch II Region

The switch 2 region is anchored by the GEF to maintain a nucleotide-releasing conformation.

Ala-59: The backbone carbonyl of Ala-59Rac1 forms a key hydrogen bond with GEFs, such as Lys-201P-Rex1 in P-Rex1 and Lys 335Vav1 in Vav1. This interaction is highly conserved and is critical for activation by promoting GDP dissociation. Gly-60: The backbone oxygen forms a hydrogen bond with GEF residues, such as His 337Vav1 in Vav1. Asp-65 and Arg-66: These are the most intimately involved residues in the Rac1-

Vav1 interface. They form extensive hydrogen bonds (seven bonds in total) with helix $\alpha 6$ of the DH domain in Vav1. They also form critical hydrogen bonds with P-Rex1 residues, such as Asn-238P-Rex1. Tyr-72: This residue is buried into a hydrophobic pocket formed by Tiam1 and Rac1 residues, anchoring the switch 2 region during Tiam1 interaction.

Other Important Residues

Ala-3: This residue is one of the six divergent residues responsible for determining DOCK specificity. Gly-54: Although its role in GEF binding is not detailed structurally in the sources, its importance is implied by mutation data; a G54A Rac mutant was unable to be activated by DOCK1DHR2-C. Trp-56: This is a key determinant of selectivity across both DOCK and DH-PH GEF families. In Rac1, Trp-56 is a bulky residue. It inserts into a pocket formed by GEF residues (like those in DOCK2's lobe C), and its presence generally supports activation by Rac-specific DOCKs. Replacing Trp-56 with Phe (as found in Cdc42) prevents optimal contact with DOCK2. P-loop (Residues 10–18): This region coordinates the binding to the phosphate groups of the nucleotide but generally does not undergo major conformational changes or direct contacts with GEFs like P-Rex1.

In essence, GEF binding uses highly conserved residues within the switch regions to remodel Rac1's conformation, facilitating nucleotide release, while differences in a few specific residues, notably Ala-27 and Trp-56, dictate whether a GEF is Rac-specific or Cdc42-specific.

Conclusion

Acknowledgement

Supporting Information Available

List supplementary figures here.

References

(1) Toma-Fukai, S.; Shimizu, T. Structural insights into the regulation mechanism of small GTPases by GEFs. *Molecules* **2019**, *24*, 3308.

(2) Cherfils, J.; Zeghouf, M. Regulation of small GTPases by GEFs, GAPs, and GDIs. *Physiological reviews* **2013**, *93*, 269–309.

(3) Loirand, G.; Sauzeau, V.; Pacaud, P. Small G proteins in the cardiovascular system: physiological and pathological aspects. *Physiological reviews* **2013**, *93*, 1659–1720.

(4) Cherfils, J.; Chardin, P. GEFs: structural basis for their activation of small GTP-binding proteins. *Trends in biochemical sciences* **1999**, *24*, 306–311.

(5) Cherfils, J. GEFs and GAPs: mechanisms and structures. *Ras Superfamily Small G Proteins: Biology and Mechanisms 1: General Features, Signaling* **2014**, 51–63.

(6) Bos, J. L.; Rehmann, H.; Wittinghofer, A. GEFs and GAPs: critical elements in the control of small G proteins. *Cell* **2007**, *129*, 865–877.

(7) DerMardirossian, C.; Bokoch, G. M. GDIs: central regulatory molecules in Rho GTPase activation. *Trends in cell biology* **2005**, *15*, 356–363.

(8) Maruta, H. In *G Proteins, Cytoskeleton and Cancer*; Maruta, H., Kohama, K., Eds.; R.G. LANDES COMPANY: Austin, Texas, USA, 1998; pp 151–170.

(9) Acuner, S. E.; Sumbul, F.; Torun, H.; Haliloglu, T. Oncogenic mutations on Rac1 affect global intrinsic dynamics underlying GTP and PAK1 binding. *Biophysical journal* **2021**, *120*, 866–876.

(10) Gray, J. L.; von Delft, F.; Brennan, P. E. Targeting the small GTPase superfamily through their regulatory proteins. *Angewandte Chemie International Edition* **2020**, *59*, 6342–6366.

(11) Traut, T. W. Physiological concentrations of purines and pyrimidines. *Molecular and cellular biochemistry* **1994**, *140*, 1–22.

(12) Pettersen, E. F.; Goddard, T. D.; Huang, C. C.; Couch, G. S.; Greenblatt, D. M.; Meng, E. C.; Ferrin, T. E. UCSF Chimera—a visualization system for exploratory research and analysis. *Journal of computational chemistry* **2004**, *25*, 1605–1612.

(13) Fiser, A.; Šali, A. *Methods in enzymology*; 2003; Vol. 374; pp 461–491.

(14) Yang, Z.; Lasker, K.; Schneidman-Duhovny, D.; Webb, B.; Huang, C. C.; Pettersen, E. F.; Goddard, T. D.; Meng, E. C.; Sali, A.; Ferrin, T. E. UCSF Chimera, MODELLER, and IMP: an integrated modeling system. *Journal of structural biology* **2012**, *179*, 269–278.

(15) Huang, C. C.; Meng, E. C.; Morris, J. H.; Pettersen, E. F.; Ferrin, T. E. Enhancing UCSF Chimera through web services. *Nucleic acids research* **2014**, *42*, W478–W484.

(16) Würtele, M.; Wolf, E.; Pederson, K. J.; Buchwald, G.; Ahmadian, M. R.; Barbieri, J. T.; Wittinghofer, A. How the Pseudomonas aeruginosa ExoS toxin downregulates Rac. *Nature structural biology* **2001**, *8*, 23–26.

(17) Krauthammer, M.; Kong, Y.; Ha, B. H.; Evans, P.; Bacchiocchi, A.; McCusker, J. P.; Cheng, E.; Davis, M. J.; Goh, G.; Choi, M.; others Exome sequencing identifies recurrent somatic RAC1 mutations in melanoma. *Nature genetics* **2012**, *44*, 1006–1014.

(18) Lucato, C. M.; Halls, M. L.; Ooms, L. M.; Liu, H.-J.; Mitchell, C. A.; Whisstock, J. C.; Ellisdon, A. M. The phosphatidylinositol (3, 4, 5)-trisphosphate-dependent Rac exchanger 1· Ras-related C3 botulinum toxin substrate 1 (P-Rex1· Rac1) complex reveals the basis of Rac1 activation in breast cancer cells. *Journal of Biological Chemistry* **2015**, *290*, 20827–20840.

(19) Worthylake, D. K.; Rossman, K. L.; Sondek, J. Crystal structure of Rac1 in complex with the guanine nucleotide exchange region of Tiam1. *Nature* **2000**, *408*, 682–688.

(20) Rapley, J.; Tybulewicz, V. L.; Rittinger, K. Crucial structural role for the PH and C1 domains of the Vav1 exchange factor. *EMBO reports* **2008**, *9*, 655–661.

(21) Kulkarni, K.; Yang, J.; Zhang, Z.; Barford, D. Multiple factors confer specific Cdc42 and Rac protein activation by dedicator of cytokinesis (DOCK) nucleotide exchange factors. *Journal of Biological Chemistry* **2011**, *286*, 25341–25351.

(22) Maier, J. A.; Martinez, C.; Kasavajhala, K.; Wickstrom, L.; Hauser, K. E.; Simmerling, C. ff14SB: improving the accuracy of protein side chain and backbone parameters from ff99SB. *Journal of chemical theory and computation* **2015**, *11*, 3696–3713.

(23) Peters, M. B.; Yang, Y.; Wang, B.; Fusti-Molnar, L.; Weaver, M. N.; Merz Jr, K. M. Structural survey of zinc-containing proteins and development of the zinc AMBER force field (ZAFF). *Journal of chemical theory and computation* **2010**, *6*, 2935–2947.

(24) Jorgensen, W. L.; Chandrasekhar, J.; Madura, J. D.; Impey, R. W.; Klein, M. L. Comparison of simple potential functions for simulating liquid water. *The Journal of chemical physics* **1983**, *79*, 926–935.

(25) Essmann, U.; Perera, L.; Berkowitz, M. L.; Darden, T.; Lee, H.; Pedersen, L. G. A smooth particle mesh Ewald method. *The Journal of chemical physics* **1995**, *103*, 8577–8593.

(26) Götz, A. W.; Williamson, M. J.; Xu, D.; Poole, D.; Le Grand, S.; Walker, R. C. Routine microsecond molecular dynamics simulations with AMBER on GPUs. 1. Generalized born. *Journal of chemical theory and computation* **2012**, *8*, 1542.

(27) Salomon-Ferrer, R.; Gotz, A. W.; Poole, D.; Le Grand, S.; Walker, R. C. Routine microsecond molecular dynamics simulations with AMBER on GPUs. 2. Explicit solvent particle mesh Ewald. *Journal of chemical theory and computation* **2013**, *9*, 3878–3888.

(28) Le Grand, S.; Götz, A. W.; Walker, R. C. SPFP: Speed without compromise—A mixed precision model for GPU accelerated molecular dynamics simulations. *Computer Physics Communications* **2013**, *184*, 374–380.

(29) Case, D. A.; Aktulga, H. M.; Belfon, K.; Ben-Shalom, I.; Brozell, S. R.; Cerutti, D. S.; Cheatham III, T. E.; Cruzeiro, V. W. D.; Darden, T. A.; Duke, R. E.; others *Amber 2022*; University of California, San Francisco, 2022.

(30) Case, D. A.; Cheatham III, T. E.; Darden, T.; Gohlke, H.; Luo, R.; Merz Jr, K. M.; Onufriev, A.; Simmerling, C.; Wang, B.; Woods, R. J. The Amber biomolecular simulation programs. *Journal of computational chemistry* **2005**, *26*, 1668–1688.

(31) Ryckaert, J.-P.; Ciccotti, G.; Berendsen, H. J. Numerical integration of the cartesian equations of motion of a system with constraints: molecular dynamics of n-alkanes. *Journal of computational physics* **1977**, *23*, 327–341.

(32) Roe, D. R.; Cheatham III, T. E. PTRAJ and CPPTRAJ: software for processing and analysis of molecular dynamics trajectory data. *Journal of chemical theory and computation* **2013**, *9*, 3084–3095.

(33) Durrant, J. D.; de Oliveira, C. A. F.; McCammon, J. A. POVME: an algorithm for measuring binding-pocket volumes. *Journal of Molecular Graphics and Modelling* **2011**, *29*, 773–776.

(34) Durrant, J. D.; Votapka, L.; Sørensen, J.; Amaro, R. E. POVME 2.0: an enhanced tool for determining pocket shape and volume characteristics. *Journal of chemical theory and computation* **2014**, *10*, 5047–5056.

(35) Pérot, S.; Sperandio, O.; Miteva, M. A.; Camproux, A.-C.; Villoutreix, B. O. Druggable pockets and binding site centric chemical space: a paradigm shift in drug discovery. *Drug discovery today* **2010**, *15*, 656–667.

(36) Nayal, M.; Honig, B. On the nature of cavities on protein surfaces: application to the identification of drug-binding sites. *Proteins: Structure, Function, and Bioinformatics* **2006**, *63*, 892–906.

(37) An, J.; Totrov, M.; Abagyan, R. Pocketome via comprehensive identification and classification of ligand binding envelopes. *Molecular & Cellular Proteomics* **2005**, *4*, 752–761.

(38) Martínez, L.; Andrade, R.; Birgin, E. G.; Martínez, J. M. PACKMOL: A package for building initial configurations for molecular dynamics simulations. *Journal of computational chemistry* **2009**, *30*, 2157–2164.

(39) Grant, B. J.; Rodrigues, A. P.; ElSawy, K. M.; McCammon, J. A.; Caves, L. S. Bio3d: an R package for the comparative analysis of protein structures. *Bioinformatics* **2006**, *22*, 2695–2696.

(40) Miller III, B. R.; McGee Jr, T. D.; Swails, J. M.; Homeyer, N.; Gohlke, H.; Roitberg, A. E. MMPBSA. py: an efficient program for end-state free energy calculations. *Journal of chemical theory and computation* **2012**, *8*, 3314–3321.

(41) Onufriev, A.; Bashford, D.; Case, D. A. Exploring protein native states and large-scale conformational changes with a modified generalized born model. *Proteins: Structure, Function, and Bioinformatics* **2004**, *55*, 383–394.

(42) Weiser, J.; Shenkin, P. S.; Still, W. C. Approximate atomic surfaces from linear combinations of pairwise overlaps (LCPO). *Journal of Computational Chemistry* **1999**, *20*, 217–230.

(43) Sun, Q.; Burke, J. P.; Phan, J.; Burns, M. C.; Olejniczak, E. T.; Waterson, A. G.; Lee, T.; Rossanese, O. W.; Fesik, S. W. Discovery of small molecules that bind to K-Ras and inhibit Sos-mediated activation. *Angewandte Chemie International Edition* **2012**, *51*, 6140–6143.

(44) Maurer, T.; Garrenton, L. S.; Oh, A.; Pitts, K.; Anderson, D. J.; Skelton, N. J.; Fauber, B. P.; Pan, B.; Malek, S.; Stokoe, D.; others Small-molecule ligands bind to a distinct pocket in Ras and inhibit SOS-mediated nucleotide exchange activity. *Proceedings of the National Academy of Sciences* **2012**, *109*, 5299–5304, Publisher: National Academy of Sciences.

(45) Kukimoto-Niino, M.; Katsura, K.; Kaushik, R.; Ehara, H.; Yokoyama, T.; Uchikubo-Kamo, T.; Nakagawa, R.; Mishima-Tsumagari, C.; Yonemochi, M.; Ikeda, M.; others

Cryo-EM structure of the human ELMO1-DOCK5-Rac1 complex. *Science advances* **2021**, *7*, eabg3147.

(46) Bruciaferri, N.; Eberhardt, J.; Llanos, M. A.; Loeffler, J. R.; Holcomb, M.; Fernandez-Quintero, M. L.; Santos-Martins, D.; Ward, A. B.; Forli, S. CosolvKit: a versatile tool for cosolvent MD preparation and analysis. *Journal of Chemical Information and Modeling* **2024**, *64*, 8227–8235.